# Supplementary Material

## NEURONPYXL SPREADSHEET TUTORIAL

### Biophysical Properties of Cells

Each cell in NEURONpyxl is treated as a point cell, and all units are converted from NEURON's distributed units to absolute units to match the values reported in SNNAP. Therefore, each cell is composed of only one compartment and the axial resistance, length and diameter of each cell are insignificant quantities and are hence left at their default values (Carnevale and Hines, 2006). There is also no temperature dependence. Future work could implement multi-compartment models for NEURONpyxl cells.

To add a cell, write its name in the left-most column of the region shown in Figure S1. You must also enter the membrane capacitance in $\mu$F. All of the other sections of the spreadsheet will update when cells are added to the network.

### Ion Channels

In the "Neu" sheet, you can add neurons and edit parameters for the ion channels, shown also in Figure S1. The ion channels determine which ions within NEURON are used. In NEURON, you can use either $Ca^{2+}$, $Na^{+}$, $Cl^{-}$, $K^{+}$ or a nonspecific current. To use either of the four ions, the name of your ion channel must start with one of the four ions (e.g. Ka, Napp, Cl, CaN). If the name of the ion channel does not start with an ion, then NEURON will use a nonspecific current. Ion channel names must start with a letter.

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | AA | AB | AC | AD | AE | AF | AG |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | leak | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 2 | | | | File | vdg | | | vdg | | | | | | | A | | | | | | | | | | | | | B | | | | | |
| 3 | | Name | Template | cm | g | E | g | p | E | An | h | s | p | tmx | tmin | th1 | ts1 | tp1 | th2 | ts2 | tp2 | Bn | h | s | p | tmx | tmin | th1 | ts1 | tp1 | th2 | ts2 | tp2 |
| 4 | B4 | B4 | | 0.007 | 0.1 | -55 | 60 | 3 | 30 | 0 | -28.5 | 8.6 | 1 | 0.0006 | 1E-04 | -40.5 | 6 | 1 | | | | 0 | -38 | 5 | 1 | 0.005 | 0.0002 | -37 | 11.6 | 1 | | | |
| 5 | B8 | B8 | | 0.01 | 0.09 | -59 | 25 | 3 | 30 | 0 | -37 | 5 | 1 | 0.006 | 0.001 | -43 | 4 | 1 | | | | 0 | -43 | 5 | 1 | 0.02 | 0.005 | -38 | 10 | 1 | | | |
| 6 | B20 | B20 | | 0.01 | 0.05 | -60 | 28 | 3 | 30 | 0 | -39 | 5 | 1 | 0.006 | 0.001 | -45 | 4 | 1 | | | | 0 | -45 | 5 | 1 | 0.02 | 0.005 | -40 | 10 | 1 | | | |
| 7 | B30 | B30 | | 0.01 | 0.1 | -55 | 25 | 3 | 30 | 0 | -37 | 5 | 1 | 0.006 | 0.001 | -43 | 4 | 1 | | | | 0 | -43 | 5 | 1 | 0.02 | 0.005 | -38 | 10 | 1 | | | |
| 8 | B31s | B31s | | 0.01 | 0.05 | -60 | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 9 | B31a | B31a | | 0.01 | 0.04 | -57 | 25 | 3 | 30 | 0 | -37 | 5 | 1 | 0.006 | 0.001 | -43 | 4 | 1 | | | | 0 | -43 | 5 | 1 | 0.02 | 0.005 | -38 | 10 | 1 | | | |
| 10 | B34 | B34 | | 0.01 | 0.09 | -59 | 25 | 3 | 30 | 0 | -39 | 5 | 1 | 0.006 | 0.001 | -45 | 4 | 1 | | | | 0 | -45 | 5 | 1 | 0.02 | 0.005 | -40 | 10 | 1 | | | |
| 11 | B35 | B35 | | 0.0005 | 0.1 | -60 | 7.5 | 3 | 50 | 0 | -39 | 5 | 1 | 0.003 | 9E-04 | -40 | 2 | 1 | | | | 0 | -49 | 9 | 1 | 0.02 | 0.0048 | -36 | 3.5 | 1 | | | |
| 12 | B40 | B40 | | 0.01 | 0.1 | -57 | 25 | 3 | 30 | 0 | -37 | 5 | 1 | 0.006 | 0.001 | -43 | 4 | 1 | | | | 0 | -43 | 5 | 1 | 0.02 | 0.005 | -38 | 10 | 1 | | | |
| 13 | B51s | B51s | | 0.001 | 0.15 | -64 | 3 | 3 | 30 | 0 | -37 | 5 | 1 | 0.006 | 0.001 | -43 | 4 | 1 | | | | 0 | -43 | 5 | 1 | 0.02 | 0.005 | -38 | 10 | 1 | | | |
| 14 | B51a | B51a | | 0.003 | 0.16 | -77 | 25 | 3 | 30 | 0 | -37 | 5 | 1 | 0.006 | 0.001 | -43 | 4 | 1 | | | | 0 | -43 | 5 | 1 | 0.02 | 0.005 | -38 | 10 | 1 | | | |
| 15 | B52 | B52 | | 0.01 | 0.1 | -62 | 25 | 3 | 30 | 0 | -37 | 5 | 1 | 0.006 | 0.001 | -43 | 4 | 1 | | | | 0 | -43 | 5 | 1 | 0.02 | 0.005 | -38 | 10 | 1 | | | |
| 16 | B63 | B63 | | 0.01 | 0.04 | -60 | 25 | 3 | 30 | 0 | -37 | 5 | 1 | 0.006 | 0.001 | -43 | 4 | 1 | | | | 0 | -43 | 5 | 1 | 0.02 | 0.005 | -38 | 10 | 1 | | | |
| 17 | B64s | B64s | | 0.01 | 0.08 | -60 | 2.5 | 3 | 30 | 0 | -37 | 5 | 1 | 0.006 | 0.001 | -43 | 4 | 1 | | | | 0 | -43 | 5 | 1 | 0.02 | 0.005 | -38 | 10 | 1 | | | |
| 18 | B64a | B64a | | 0.01 | 0.1 | -60 | 25 | 3 | 30 | 0 | -37 | 5 | 1 | 0.006 | 0.001 | -43 | 4 | 1 | | | | 0 | -43 | 5 | 1 | 0.02 | 0.005 | -38 | 10 | 1 | | | |
| 19 | B65 | B65 | | 0.01 | 0.08 | -55 | 25 | 3 | 30 | 0 | -37 | 5 | 1 | 0.006 | 0.001 | -43 | 4 | 1 | | | | 0 | -43 | 5 | 1 | 0.02 | 0.005 | -38 | 10 | 1 | | | |
| 20 | CBI2 | CBI2 | | 0.01 | 0.1 | -60 | 25 | 3 | 30 | 0 | -37 | 5 | 1 | 0.006 | 0.001 | -43 | 4 | 1 | | | | 0 | -43 | 5 | 1 | 0.02 | 0.005 | -38 | 10 | 1 | | | |
| 21 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 22 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 23 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 24 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

**Figure S1.** Table in the "Neu" sheet where cells are added to the network, membrane capacitance is defined, and ion channels and their parameters are defined.

**Conductance to ion**

| Name | ch | ion | ch | ion | ch | ion | ch | ion |
|---|---|---|---|---|---|---|---|---|
| B4 | CaN | Ca | | | | | | |
| B8 | | | | | | | | |
| B20 | Na | Na | | | | | | |
| B30 | Na | Na | | | | | | |
| B31s | | | | | | | | |
| B31a | Na | Na | | | | | | |
| B34 | Na | Na | | | | | | |
| B35 | | | | | | | | |
| B40 | | | | | | | | |
| B51s | | | | | | | | |
| B51a | | | | | | | | |
| B52 | | | | | | | | |
| B63 | Na | Na | | | | | | |
| B64s | | | | | | | | |
| B64a | | | | | | | | |
| B65 | Na | Na | | | | | | |
| CBI2 | Na | Na | | | | | | |

**Ion pools**

| Name | ion | K1 | K2 | ion | K1 | K2 | ion | K1 | K2 |
|---|---|---|---|---|---|---|---|---|---|
| B4 | Ca | 10 | 9 | | | | | | |
| B8 | | | | | | | | | |
| B20 | Na | 1 | 0.2 | | | | | | |
| B30 | Na | 1 | 0.2 | | | | | | |
| B31s | | | | | | | | | |
| B31a | Na | 14 | 0.4 | | | | | | |
| B34 | Na | 1 | 0.2 | | | | | | |
| B35 | | | | | | | | | |
| B40 | | | | | | | | | |
| B51s | | | | | | | | | |
| B51a | | | | | | | | | |
| B52 | | | | | | | | | |
| B63 | Na | 14 | 0.4 | | | | | | |
| B64s | | | | | | | | | |
| B64a | | | | | | | | | |
| B65 | Na | 1 | 0.2 | | | | | | |
| CBI2 | Na | 1 | 0.2 | | | | | | |

**Ion to conductance**

| Name | ion | ch | opt1 | opt2 | p1 | p2 | b | ion | ch | opt1 | opt2 | p1 | p2 | b |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B4 | Ca | CaN | 3 | 2 | 60 | | 17 | Ca | Kcaf | 1 | 1 | 0.01 | | |

**Figure S2.** Tables in the "Neu" sheet where ion pools and ion pool regulation are defined.

A leak channel is also included by default. If you do not want to use it, simply set the conductance to 0. Lastly, leave the "Template" column empty.

Each ion channel has a name, and parameters for an $A$ and a $B$ component, corresponding to equations 15, 16, 17, and 18. "vdg" stands for voltage-dependent conductance. Under "vdg", you must enter values for g (conductance), p (power), and E (reversal potential), corresponding to equation 14. For $A$ (activation) and $B$ (inactivation) you must fill in at least An/Bn (initial value), h, s, p, and tmx. If you only enter these values, then equation 20 will be used. If you also enter values for "th1", "ts1", and "tp1", then equation 15 ($A$) or 17 ($B$) will be used. If you additionally fill in "th2", "ts2", and "tp2", then equation 16 ($A$) or 18 ($B$) will be used.

## Ion Pools

Ion pools can also be added in the "Neu" sheet using the section shown in the "Ion pools" table in Figure S2. Enter the type of ion of the ion pool in the "ion" column. The ion pool will update the intracellular concentration in NEURON. The columns "K1" and "K2" correspond to the parameters in equation 22.

The ion pools must be supplied by at least one ion channel. This is defined in the "Conductance to ion" table shown in Figure S2. There can be multiple ions that feed a single pool (given by the "ch" column), and an ion pool ("ion" column) requires at least one ion channel to supply it.

To regulate an ion channel with an ion pool, enter parameters in the "Ion to conductance" table shown in Figure S2. The "ion" column is which ion pool will regulate the channel in the "ch" column. The number entered into the "opt1" column defines which of the functions in equation 23 to use. The number entered into the "opt2" column defines which of the functions in equation 24 to use. Provide values of "p1", "p2",

| Presynaptic \ Postsynaptic | B4 | B8 | B20 | B30 | B31s | B31a | B34 | B35 | B40 | B51s | B51a | B52 | B63 | B64s | B64a | B65 | CBI2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B4 | | | | | 0.0086 | | | | | | 0.09 | | | 0.014 | | | |
| B8 | | | | | | | | | | | | | | | | | |
| B20 | | | | | 0.004 | | | | | | | | 0.004 | | | 0.004 | |
| B30 | | | | | 0.009 | | | | | | | | 0.006 | | | | |
| B31s | 0.0082 | | 0.004 | 0.009 | | 0.012 | 0.008 | 0.0075 | | | | | 0.015 | | | 0.004 | |
| B31a | | | | | 0.012 | | | | | | | | | | | | |
| B34 | | | | | 0.008 | | | | 0.008 | | | | | | | 0.008 | |
| B35 | | | | | 0.0075 | | | | | | | | | | | | |
| B40 | | | | | | | 0.008 | | | | | | 0.004 | | | 0.008 | |
| B51s | | | | | | | | | | | 0.2 | | | 0.038 | | | |
| B51a | 0.135 | | | | | | | | | 0.2 | | | | | 0.02 | | |
| B52 | | | | | | | | | | | | | | | | | |
| B63 | | | 0.004 | 0.006 | 0.015 | | | | 0.004 | | | | | | | 0.006 | |
| B64s | 0.034 | | | | | | | | | 0.12 | | | | | 0.25 | | |
| B64a | | | | | | | | | | | 0.02 | | | 0.25 | | | |
| B65 | | | 0.004 | | 0.004 | | 0.008 | | 0.008 | | | | 0.006 | | | | |
| CBI2 | | | | | | | | | | | | | | | | | |

**Figure S3.** Table in the "es" sheet where gap functions are defined.

and "b" as required by the equations you have selected and their corresponding units (see the section Implementing SNNAP in NEURON).

## Electrical Synapses

Electrical synapses are defined in the matrix shown in Figure S3. The conductances that correspond with equation 26 are entered into the matrix. It is not required to enter a conductance for both neurons in the electrical synapse. You should not connect a neuron to itself with an electrical synapse.

## Chemical Synapses

There is a similar matrix for defining chemical synapses. However, since the equations are more complex, there are now 3 matrices to fill out in order to define a complete chemical synapse. First, you must define the synaptic conductance in the "cs_g" sheet. Then, you must define the reversal potential in the "cs_E" sheet. Then you must define the synaptic activation functions in the "cs_FAT" sheet. Both the "cs_g" and "cs_E" sheets behave exactly like the "es" sheet, except there is an option for two chemical synapses within each cell of the matrix. The upper cell is called the "fast" synapse and the lower cell is called the "slow" synapse. That said, the paramter choices are still up to the user.

| Presynaptic \ Postsynaptic | B4 | B8 | B20 | B30 | B31s | B31a | B34 | B35 | B40 | B51s | B51a | B52 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B4 | | 2 | | | 2 | | | | | 36 | | 2 |
| B8 | | | | | | | | | | | | |
| B20 | 10<br>25 | 5<br>6 | | | | | | | | | | |
| B30 | | 7<br>26 | | | 8 | | | | | | | |
| B31s | | | | | | | | | | | | |
| B31a | | | | | 12 | | | | | | | |
| B34 | | 1<br>29 | 4 | | 9<br>37 | | | | | | | |
| B35 | 15<br>16 | | | | | | | | | | 2 | 17<br>18 |
| B40 | | 2<br>19 | | | 2 | | | | | | | |
| B51s | | | | | | | | | | | | |
| B51a | | 2 | | | | | | | | | | 20 |
| B52 | | 2 | | | | | | | | 20<br>4 | | |

| | | taus | | depression | | Voltage dependence | | | | facilitation | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | u1 | u2 | ud | ur | h | s | p | tx | ion | u |
| 11 | 1 | 0.01 | | | | | | | | | |
| 11 | 2 | 0.015 | | | | | | | | | |
| 3 | 3 | 0.1 | | | | | | | | | |
| 3 | 4 | 0.2 | | | | | | | | | |
| 3 | 5 | 0.004 | 0.0015 | | | | | | | Na | 0.5 |
| 2 | 6 | 1 | | | | -31 | 3 | 1 | 0.5 | Na | 0.5 |
| 1 | 7 | 0.2 | 0.001 | | | | | | | | |
| 1 | 8 | 0.005 | | | | | | | | Na | 0.5 |
| 9 | 9 | 0.01 | | | | | | | | Na | 0.5 |
| 3 | 10 | 0.02 | | | | | | | | | |
| 1 | 11 | 0.5 | 5 | | | -30 | 15 | 1 | 0.5 | | |
| 2 | 12 | 0.2 | | | | -48 | 4 | 3 | 0.1 | Na | 0.5 |
| 1 | 13 | 1.5 | | | | | | | | | |

**Figure S4.** Table in the "cs_FAT" sheet where synaptic activations are defined.

The "cs_FAT" sheet is more complex. In each cell of the matrix, enter the number that corresponds to the set of equations in the table on the right side of the sheet.

In the "taus" table, the parameters "u1" and "u2" correspond to $\tau_1$ and $\tau_2$ in equation 29. If only "u1" is provided, then $\tau_1 = \tau_2$. This is the only activation where parameters are required.

In the "depression" table, the parameters "ud" and "ur" correspond to $\tau_d$ and $\tau_r$ in equation 31. If either "ud" or "ur" are not provided, there will be no modulation.

In the "voltage dependence" table, the parameters correspond to those in equation 33. If "tmx" is not provided, it will set voltage-dependent activation to its steady state (equation 34). If any of the other parameters are not provided, there will be no voltage-dependence.

In the "faciliation" table, the parameters correspond to those in equation 25. The synaptic activation will be modulated by an ion, with a time constant of "u". If either of the parameters are not provided, there will be no modulation.

## Simulation Control

To initialize the voltages, fill in the "Initial voltage" table in the "Neu" sheet in Figure S5. Each cell should have an initialized voltage.

**Initial voltage**

| Name | mV |
|---|---|
| B4 | -60 |
| B8 | -60 |
| B20 | -60 |
| B30 | -60 |
| B31s | -60 |
| B31a | -60 |
| B34 | -60 |
| B35 | -60 |
| B40 | -60 |
| B51s | -60 |
| B51a | -60 |
| B52 | -60 |

| Current injection | | | |
|---|---|---|---|
| neuron | start | stop | magnitude |
| B4 | 0 | 30 | -2 |
| B8 | 0 | 30 | -2 |
| B20 | 0 | 30 | -2 |
| B30 | 0 | 30 | -2 |
| B31s | 0 | 30 | -2 |
| B34 | 0 | 30 | -2 |
| B35 | 0 | 30 | -2 |
| B40 | 0 | 30 | -2 |
| B51s | 0 | 30 | -2 |

**Figure S5.** Tables where voltages are initialized and current clamps are defined.

To define an external stimulation, you can apply a current clamp in a sheet which has the extension ".smu". You can have any number of current clamps, as long as the neuron, amplitude of current, the start time and the end time of the clamp are provided. Any other tables in the ".smu" sheet are for SNNAP simulations, so further simulation control must be done in the command line or the Python API. For more simulation control options, see the README.md file in the GitHub repository.